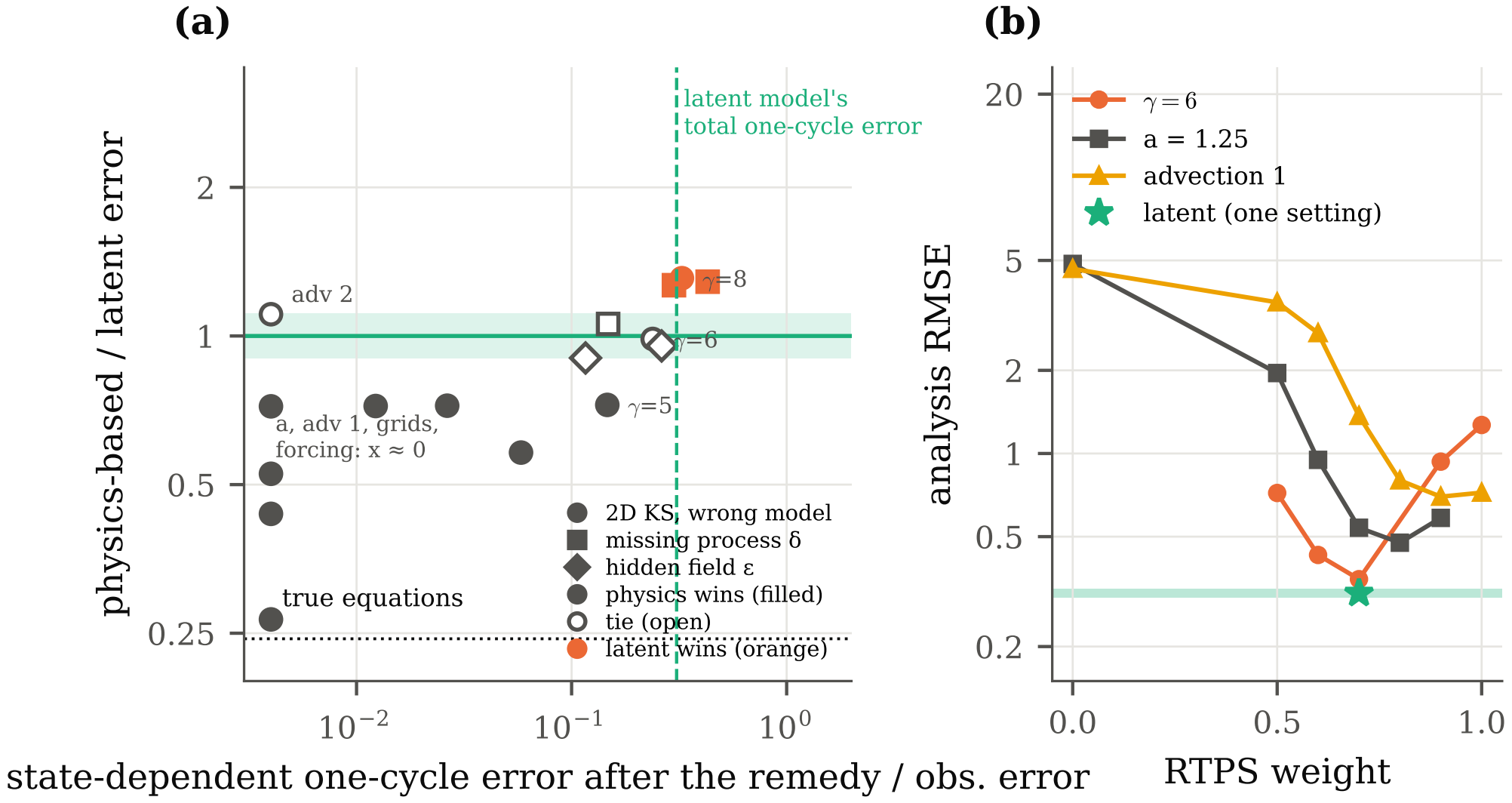
(a)
(b)
physics-based / latent error
latent model's
total one-cycle error
adv 2
γ=8
γ=6
γ=5
a, adv 1, grids,
forcing: x ≈ 0
true equations
2D KS, wrong model
missing process δ
hidden field ε
physics wins (filled)
tie (open)
latent wins (orange)
2
1
0.5
0.25
$10^{-2}$
$10^{-1}$
$10^{0}$
state-dependent one-cycle error after the remedy / obs. error
analysis RMSE
γ = 6
a = 1.25
advection 1
latent (one setting)
20
5
2
1
0.5
0.2
0.0
0.5
1.0
RTPS weight